

Applied AI for Bioimage Analysis

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Cell biologist translating wet-lab questions into validated, reproducible AI-assisted imaging workflows.

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Applied AI Bioimage Workflow



Wet-lab foundation	Applied AI	Scientific validation
Cell culture, CRISPR/Cas9, 2D/3D disease models and functional assays.	Cellpose training, Napari integration, Python environments and quantitative imaging.	Data quality, model evaluation, reproducibility and biological interpretation.

Flagship evidence: AI-assisted 3D cyst quantification

A real assay bottleneck required biological design, microscopy, machine learning, software collaboration, and laboratory implementation.

~14,000	1 week / 2 people	~1 hour
cysts per experiment	manual FIJI analysis	Cellpose-Napari analysis

Problem: Manual 3D cyst measurement limited throughput and required substantial researcher time.

Contribution: Trained Cellpose models, collaborated with Guillaume Witz at the University of Bern Data Science Lab on a Napari plugin, and set up the Python/Cellpose laboratory environment.

Implementation: Created reusable templates, protocols, and structured documentation.

Outcome: Reduced analysis from about one week of manual work by two researchers to about one hour; the framework was adapted for other models.

Confidentiality: No unpublished images, model weights, code, treatment-response data, or biological conclusions are included.

Public reproducible project

Cellpose-SAM benchmark on BBBC039 U2OS nuclei from a chemical screen

Dataset	200 Hoechst-stained U2OS fields and approximately 23,000 manually annotated nuclei.
Objective	Demonstrate public-data access, AI segmentation, validation, error analysis, and reproducible reporting.
Approach	Pretrained Cellpose-SAM baseline, validation parameter sweep, limited fine-tuning, and held-out evaluation.
Outputs	Notebook, Python modules, CSV/JSON reports, tests, CI, environment files, and public case study.

Workflow and employer relevance

- **Data & QC:** official downloads, mask decoding, intensity and object-density checks
- **AI baseline:** Cellpose-SAM inference with documented settings and runtime
- **Validation:** parameter selection on validation data; AP, AJI, IoU and count error
- **Fine-tuning:** limited training on annotated images using the current Cellpose workflow
- **Error analysis:** review failure cases by morphology, density, overlap and illumination
- **Reproducibility:** fixed seed, environment files, tests, CI, provenance and limitations

Skills evidence for current scientist roles

Requirement	Evidence
AI-enabled tools	Cellpose-SAM use, fine-tuning and responsible interpretation.
Python / quantitative analysis	Reusable modules, batch processing, QC and machine-readable outputs.
Model validation	Train/validation/test logic, AP/AJI/IoU, count error and failure analysis.
Wet-lab integration	Biological question, assay bottleneck and measurable readout.
Reproducibility	Environment files, tests, GitHub Actions CI, provenance and limitations.
Cross-functional work	Real collaboration between wet-lab research and a Data Science Lab.

Portfolio package and completion plan

The supplied repository is designed to become a shareable GitHub portfolio rather than another course certificate.

Included deliverables

- GitHub-ready repository with English README and Vietnamese setup guide.
- Google Colab/Jupyter notebook that downloads BBBC039 and runs the benchmark.
- Reusable Python utilities, unit tests, and GitHub Actions continuous integration.
- Static GitHub Pages landing page and employer-ready case studies.
- Confidentiality and data-ethics documentation, environment files, license and CITATION.cff.

Three steps before sharing

1. Upload the repository to GitHub and replace the placeholder Colab link.
2. Run the notebook in Colab; inspect success and failure examples and retain exported results.
3. Only after verification, add final AP/AJI values and a result figure to the README, PDF, CV and LinkedIn.

PROFESSIONAL POSITIONING

Wet-lab cell biologist who can identify an analysis bottleneck, collaborate with computational specialists, train and validate an existing model, and implement a reproducible AI-assisted workflow.

Recommended CV wording after completion

Applied AI Bioimage Analysis Portfolio: Built a reproducible Cellpose-SAM benchmark on a public U2OS microscopy dataset, including data quality control, validation-based parameter selection, limited model fine-tuning, AP/AJI evaluation, error analysis, unit tests, and documented Python workflows.

Selected public sources

- Broad Bioimage Benchmark Collection: BBBC039 U2OS nuclei in a chemical screen.
- Cellpose 4 documentation: application, fine-tuning and segmentation metrics.
- EMBL-EBI machine-learning and BioImage Archive training materials.
- GitHub documentation: README, CITATION.cff and Python CI best practices.